

APNAV POPTANI

apnav.poptani@gmail.com | (980) 585-6251 | linkedin.com/in/apnavp

PROFESSIONAL SUMMARY

Software developer engineer II at Amazon with 7+ years building and owning distributed systems end-to-end, from personalization pipelines feeding ML models to cross-marketplace compliance platforms spanning multiple geographies. Designs and owns entire system surfaces, not just features, consistently replacing manual processes with self-service infrastructure that compounds. Proven ability to drive cross-domain impact across e-commerce, compliance, and healthcare, and deliver impact in new domains quickly.

PROFESSIONAL EXPERIENCE

Amazon, Seattle, WA

06/2024 – Present

Software Development Engineer II

Homepage & Navigation Platform:

- Architected self-service ranking debugger with React, API Gateway, Lambda, Glue, DynamoDB, SNS, and SQS, replacing manual script investigations with async job processing, access control, and audit logging.
- Enabled engineers, stakeholders, and partner teams to self-serve personalization ranking analysis without infrastructure access, replacing manual developer investigations with a data-driven interface for real-time content results and launch validations.
- Built personalization pipeline processing daily clickstream data for AI/ML model training and real-time Datastore serving of mobile navigation tabs, measuring impact across North America.
- Identified legacy monitoring gaps as the root cause of slow incident response and built real-time CloudWatch dashboards for 3 backend services across 4 regions, upgrading observability to 1-second resolution and **cutting high-severity incident debug time by 28%**.
- Designed a content validation framework using Java and React with 6 pluggable validator types, duplicate detection, and schema enforcement surfacing real-time validation errors at submission to catch issues pre-publish, eliminating manual review bottlenecks for partner teams.
- Modernized infrastructure across 4 global regions by migrating backend authentication to cloud-native credential management and onboarding 2 microservices into the service mesh, eliminating auth-related security vulnerabilities and **improving deployment reliability by 15%**.
- Built an engineering productivity tool with historical Q&A matching and async deflection logic to triage office hour requests, **recovering 22% of SDE bandwidth previously lost to routine consultations**.
- Mentored **3 junior engineers through code reviews**, pair programming, and design discussions, with **2 receiving strong performance ratings**, fostering a culture of engineering excellence within the team.

Global Risk & Seller Compliance:

- Designed and implemented cross-marketplace sanctions screening with parent/child account identity resolution and dual-key caching for sanctions list lookups, expanding compliance to 6 new geographies and enabling regulatory launches.
- Led cross-team onboarding South Africa marketplace as technical point of contact, documented end-to-end system flows, facilitated SME design sessions, and **reduced onboarding effort from 5 to 2 engineer-weeks** by leveraging existing screening workflows.
- Built API delivering daily seller data snapshots to ML teams, powering seller risk score prediction across compliance systems.

Amazon, Seattle, WA

06/2021 – 06/2024

Software Development Engineer I

Seller Regulatory Compliance:

- Engineered identity verification and sanctions screening services for anti money laundering compliance, safeguarding seller trust across global marketplaces.
- Designed and implemented continuous monitoring for **200K+ seller accounts** with automated verification rules covering entity list ingestion, data search, case investigation, and audit management, **reducing manual compliance reviews by 32%**.
- Developed and maintained integration test coverage for compliance workflows across 6 geographies, resolving cross-region test failures and ensuring pipeline stability for sanctions screening launches.
- Built serverless data deletion pipeline using Lambda, CDK, and Step Functions, automating GDPR-compliant seller data removal across 2 storage systems in response to user deletion requests.

Healthcare / Online Pharmacy:

- Delivered pharmacy savings card feature end-to-end using REST API, Preact frontend, AJAX endpoints, and monitoring, enabling discount card access for millions of users.
- Optimized PostgreSQL queries to display pricing before geocoding location entry, expanding drug catalog visibility by 2,000+ drugs in a catalog of 60,000 and **reducing page latency by 35%**.
- Spearheaded migration of high-traffic pages from a legacy framework to Spring Boot, implementing self-triggering AJAX calls that eliminated page timeouts.
- Improved pharmacy discoverability by embedding a zip code-based interactive map for locating nearby discounted medications.
- Drove user conversion by implementing a Prime membership signup ingress into the pharmacy experience, **contributing to a 23% uplift in prime conversion rates**, built deep linking from mobile web to native app across all pharmacy pages, improving mobile engagement.
- Redesigned drug search logic by identifying a data suppression bug, re-architecting queries, and **surfacing 1,400+ previously hidden drugs** via feature-flagged A/B rollout.

Lease on the Block, New York, NY**06/2020 – 08/2020***Full-Stack Engineer Intern*

- Scraped NYC government lease data using Python, Selenium, and JavaScript to surface prior tenant issues pre-move-in, with automated validation ensuring data integrity.
- Built interactive web pages using Next.js, Material-UI, and React, drove technical architecture planning alongside the CTO.
- Designed, modified, and optimized SQL stored procedures, views, and functions for data migration with full regression testing.
- Built affordable housing billboard feature end-to-end using HTML, CSS, Python, and Flask surfacing real-time NYC rental listings for low-income residents.

Tata Consultancy Services, Pune, India**03/2017 – 07/2019***Systems Engineer*

- **Analyzed and remediated 400+ ASP.NET applications** for Morgan Stanley's Wealth Management platform, delivering comprehensive documentation and action plans.
- Built RESTful API using C#, MVC, and ASP.NET to mask sensitive PII across user interfaces, ensuring data security compliance.
- Engineered C# and ASP.NET performance tools **reducing deployment times 20%** for Morgan Stanley applications.
- Developed C# reporting app automating weekly sprint status from SharePoint, **cutting manual effort by 35%**.

CORE COMPETENCIES

Languages & Frameworks:	Java Python TypeScript JavaScript SQL Spring Boot React Preact Next.js Node.js Flask REST APIs
Cloud & Infrastructure:	AWS (Lambda, S3, DynamoDB, CloudWatch, CDK, CloudFormation, SQS, ECS Fargate, Step Functions, IAM) Infrastructure as Code Serverless Computing Microservices Architecture Event-Driven Architecture
Databases:	DynamoDB PostgreSQL MySQL MongoDB Microsoft SQL Server NoSQL SQL
DevOps & CI/CD:	CI/CD Pipelines Git Terraform Docker Kubernetes Containerization Static Analysis (Checkstyle, SpotBugs)
Software Development Best Practices:	System Design Agile Methodologies A/B Testing Integration Testing Automated Testing Frameworks JUnit Selenium Observability & Monitoring Code Reviews Unit Testing
Cross-Functional Collaboration:	Technical Documentation Stakeholder Communication Mentorship & Team Development Sprint Planning Agile Ceremonies

EDUCATION

Master of Science in Computer Science | GPA: 3.88 | 08/2019 - 05/2021*University of North Carolina at Charlotte, Charlotte, NC***Bachelor of Engineering in Computer Science | GPA: 3.70 | 06/2012 – 07/2016***Chhattisgarh Swami Vivekanand Technical University, Raipur, India*